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Improvement of "pest" oxidation resistance of Nb-Mo-W-Zr solid- solution alloy at 800 °C by gas nitridation

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ABSTRACT

Nitride layer was prepared on an Nb-5W-2Mo-1Zr by gas nitridation technique. The nitride layer was composed of an outer layer of (Nb, Me)$_4$N$_3$ and (Nb, Me)$_2$N nitrides and an inner lamellar eutectoid-like colony (LEC) interface consisting of (Nb, Me)$_2$N and Nb$_{SS}$ phases. The nitrided alloy showed parabolic oxidation kinetics curve with better oxidation resistance than that of the bare alloy when oxidized at 800°C for 100 h. The mechanisms for the nitride layer to suppress the "pest" and spallation of (Nb,W)$_x$O$_y$ oxide layer were discussed.

1. Introduction

Niobium (Nb) base alloys can be used as optimal hot-ended components material in the combustion chamber and nozzle of rocket engines due to the advantages of low density, high melting point and excellent mechanical properties at higher temperatures [1–4].With the service life of the new generation of rocket engines being prolonged to a few hour level, the oxidation resistance properties of rocket engines in the moderate temperature interval between 600 °C and 800 °C during the cruise stage is required to be higher [5,6]. However, the Nb base alloys may suffer catastrophic oxidation "pest" in this temperature range [7–9]. Therefore, improving "pest" oxidation resistance at moderate temperatures has become critical issue for the implication of the Nb base alloys.

Generally, there are two main approaches to solve this problem. The first one is to add active elements (such as Al, Ti, Hf, ect.) to the Nb base alloys. However, this strategy just plays a limit role in improving the oxidation resistance [10,11], and sometimes degrades the mechanical properties of Nb [12,13]. For instance, Nb-22Ti-16Si-2Hf-2Al-xCr samples exhibited better oxidation resistance at 1250 °C when Cr, the most beneficial element to oxidation resistance through formation of Cr_2Nb intermetallic compound, increased from 2 at.% to 17 at.%, whereas the room temperature fracture toughness decreased from 14.4 MPa $m^{1/2}$ to 10.3 MPa $m^{1/2}$ due to increase of the brittle phase fraction, and the compressive strength at 1350 °C decreased from 300 MPa to 80 MPa [12] because of a decrease in the melting point of the alloy by the addition of Cr. Another method is to design the oxidation-resistant coatings (such as silicide and aluminide coatings) to protect Nb alloys from oxidation at elevated temperatures [14,15]. Aluminide coating can form a dense Al_2O_3 oxide scale in atmosphere at high temperatures to prevent further oxidation of substrate [16]. Unfortunately, the aluminide coatings have poor mechanical properties at high temperature and are easy to spallation at inter-diffusion layer [17]. This may expose the substrate in air and result in quick "pest" of the substrate, which limits the applications of aluminide coatings. The silicide coatings have been extensively studied for their strong bonding with substrate and self-healing behaviours [9,18]. It was reported that the Nb-Si based alloy with Mo-62Si-5B coating showed a mass gain of 0.41 mg/cm^2 after oxidation at 1250 °C for 100 h [14]. However, the silicide coatings are still brittle at room temperature [19] and also show "pest" phenomenon at the moderate temperature [20], which means that the silicides are unsuitable for being protective coatings on the Nb base alloys [21]. Therefore, it is necessary to find a new coating to protect Nb from "pest" failure at the moderate temperatures.

Among various coating techniques, nitridation is commonly employed to improve the surface hardness, wear resistance, as well as oxidation resistance of metals [22–26]. It was reported that the nitridation increased the oxidation resistance of 304 steel at 800 °C because of the rapid formation of protective Cr_2O_3 oxide scale in the initial stage of oxidation [27]. Moreover, the nitride layer promoted the oxidation resistance of pure Ti, Ti-6Al-4 V (at.%) and Ti-7Nb (at.%) alloys, while the improvement degree depended on the alloy composition and the

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nitride layer thickness formed on the substrate [28]. Oppositely, the nitridation was found to accelerate the oxidation rate of a Ti-47Al-2Nb-2Cr-0.2Si(at.%) alloy at 900 °C and 1000 °C, in comparison to the un-nitrided alloys due to the limited thermo-dynamical stability of the Ti-N compound at high temperature [29]. Thus, effect of the nitridation on the oxidation resistance of alloys has some confuse conclusions and should be needed to further investigate.

It was reported that during the nitridation of Nb-10Ti alloy at 1300 °C, three different reaction layers of δ-(Nb,Ti)N_{1-x}, β-(Nb,Ti)N and β-(Nb,Ti)N+β-(Nb,Ti) lamellae from the outside in formed on the substrate [30]. Bare Nb even shown better oxidation resistance after being treated by high temperature nitrogen plasma based ion implantation; however, the mechanism of oxidation of the nitrided Nb wasn't revealed [31]. To date, few investigations with regard to improving the "pest" oxidation resistance of the Nb base alloys by nitridation have been reported. In this work, Nb-5W-2Mo-1Zr (at.%) alloy named Nb521 was treated by gas nitriding treatment to obtain a homogeneous and dense nitride layer. A comparative investigation on the microstructure of the nitride layer and the isothermal oxidation behaviours of the bare and nitrided Nb521 samples at moderate temperature (800 °C) was performed. Additionally, the mechanism for the nitride layer to resist the "pest" oxidation was analyzed.

2. Experimental procedures

2.1. Nitridation processing

The rolled commercial Nb521 alloy with a nominal composition (at.%) of Nb-5W-2Mo-1Zr was selected as substrate alloy. The fan-shaped samples, 10 mm in radius, 3 mm in height and the central angle is 90 degrees, cut from Nb521 bar, were mechanically ground using SiC paper (1500-grit) with water and cleaned ultrasonically in ethanol before nitridation. The nitridation processing was conducted in a tube furnace filled with flowing atmosphere consisting of nitrogen, argon and hydrogen, in which hydrogen and argon were used to consume residual oxygen in the furnace and adjust the atmosphere ratio, respectively. To control the proportion of atmosphere in the furnace, the two ends of the tubular furnace were connected with air passages and the flow rates of the atmosphere were controlled by three flow meters, respectively. During the entire nitriding process, the flow rate was 1 L/min for nitrogen, 0.5 L/min for argon and 0.1 L/min for hydrogen. Before the nitridation processing, the samples were placed on an alumina crucible with rectangular holes drilled at the bottom to allow the samples to stand in the trough so that the same nitriding degree could be obtained on both sides of the samples. Then, the combination of crucible and samples was placed in the furnace and a vacuum pump was used to pump the furnace to negative pressure (8 Pa), followed by the flow gas (gas composition N_2+31.25 %Ar+6.25 %H_2) was injected. When the gas pressure in the furnace exceeded atmospheric pressure, the furnace outlet valve was opened, so that the gas in the furnace maintained circulation. The samples were nitrided at 1200 °C for 12 h, and finally with furnace cooling to room temperature.

2.2. Oxidation tests

The bare and nitrided samples were isothermally oxidized at 800 °C for 100 h using an open-ended tube furnace in air. After cleaned ultrasonically in ethanol, the samples were held in an alumina crucible and placed in the uniform hot zone of the furnace so that the total weight change dependent on the oxidation time could be obtained. The weight gains of the oxidation samples were measured at the intervals of 1, 2, 3, 4, 5, 10, 20, 50 and 100 h using an analytical balance with an accuracy of 10^{-5} g (Model CPA225D, Germany). Three samples were used at each time to get the average weight gains value. In order to investigate the mechanism of the nitride layer impeding oxidation and the structural evolution of the oxide layer, five additional samples of each bare and nitrided alloy were isothermally static oxidized at 800 °C for 1, 2, 5, 20, 50 h, respectively.

2.3. Characterizations

The bare and nitrided samples, as well as the oxidized samples, were mounted in cold-setting epoxy resin followed by ground to 2000-grit SiC paper with water and polished with 50 nm aqueous silica colloidal suspension. X-ray diffraction (XRD, Model D/M-2500PC Rigaku, Japan) using Cu Kα radiation was used to identify phase constitutions of the samples under different states. Microstructural evolution of the nitride layer and oxide products were determined by field emission scanning electron microscopy (SEM, Model 200 F, Holland) with Energy Dispersive Spectroscopy (EDS). The element distributions in the cross-sections of the nitrided samples before and after oxidation were investigated by using an Electron Probe Micro Analysis (EPMA, Model JXA-8530 F, Japan) with wave dispersive spectroscopy (WDS).

To further observe the microstructures of the nitride layer and the oxide products formed on the substrate during the nitridation and oxidation and accurately identify their specific phase structures, the regions of interest of the samples were made into Transmission electron microscopy (TEM) samples. The TEM samples of two areas of interest (see yellow boxes in Figs. 5 and 14) were prepared using a focused ion beam (FIB, TESCAN LYRA 3, Czech Republic). After the interesting areas were processed by ion beam, a miniature mechanical arm moved them to a copper grid and fixed it with platinum, and then the samples were further milled by ion beam to remove the surface amorphous layer. The obtained thinning samples were analyzed using a field emission transmission electron microscopy (FEI TECNAI G^2F20, America) with Oxford INCA EDS detector.

3. Results

3.1. Phase constitution and microstructure of the nitride layer

Fig. 1 shows the XRD patterns of the Nb521 and the nitrided Nb521 samples. As shown in the picture, the characteristic diffraction peaks of Nb_2N (PDF#40-1274) and Nb_4N_3 (PDF#20-0803) phases were detected in the nitride sample, suggesting the nitride layer containing the Nb_2N and Nb_4N_3 nitrides; meanwhile, the Nb solid solution (Nb_{SS}) was the main phase of the Nb521 alloy. The surface morphologies of the Nb521 alloy before and after nitridation at 1200 °C for 12 h shown in Fig. 2 revealed that for the nitrided alloy many dispersed Nb_2N particles formed on the Nb_4N_3 nitride (Fig. 2a), while the bare alloy showed a single Nb_{SS} strip-like microstructure with evident grain boundaries along the rolling direction (Fig. 2b). The cross-sectional BSE images of the nitrided alloy shown in Fig. 3 confirmed that a homogeneous nitride

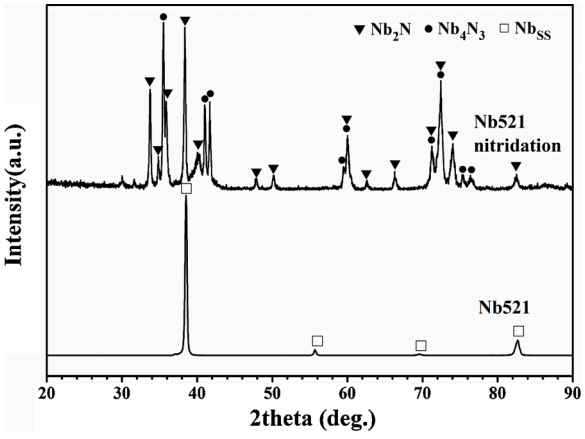


Fig. 1. X-ray diffraction patterns of the Nb521 and the nitrided Nb521 samples.

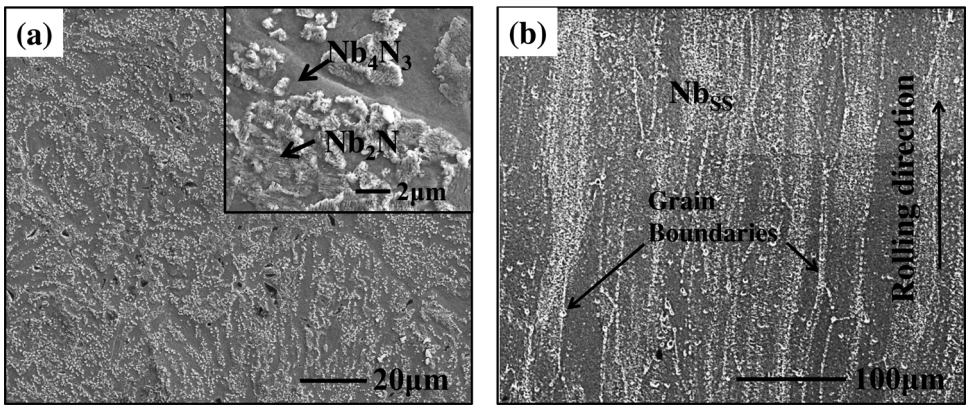


Fig. 2. Surface morphologies of the nitrided Nb521 alloy (a) and bare Nb521 alloy (b).

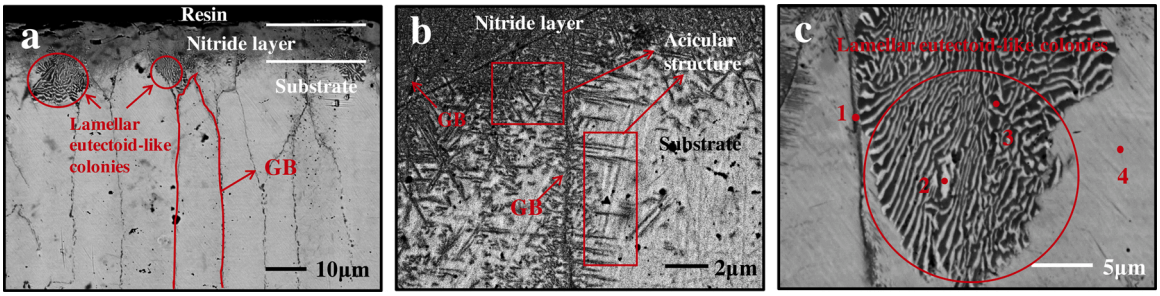
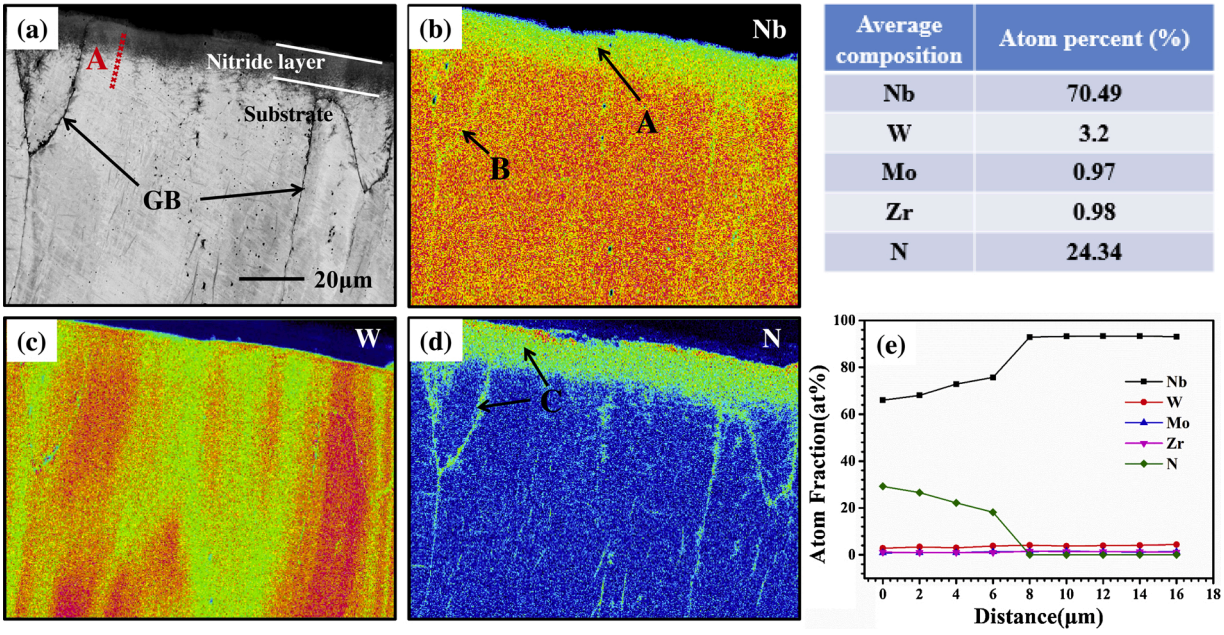


Fig. 3. Backscattered electron (BSE) images of cross-sectional morphologies of the nitrided Nb521 alloy (a) nitride layer, (b) acicular structure and (c) lamellar eutectoid-like colony.

layer covered on the substrate surface with a thickness of about 10 μm, while it adhered well to all region of the substrate (Fig. 3a). As can be seen in Fig. 3b, many needle-like strips grew on the interface between the nitride layer and the substrate or extend from the grain boundaries by degrees of approximately 60~90 toward the grain interiors at the same time. Careful investigation confirmed that lamellar eutectoid-like colonies (circles in Fig. 3a, referred as to LEC, hereafter) arisen on the interface, and two phases obviously existed in the LEC zones, as shown in an enlarged image in Fig. 3c. So, the LEC colonies represent for the interface between the nitride layer and the substrate. To identify the elemental distribution in the nitride layer, EPMA elemental maps of cross-sections of the nitride layer were detected, which are shown in Fig. 4. The elemental distributions revealed that the nitride layer (arrow A in Fig. 4b) and the grain boundaries (arrow B in Fig. 4b) depleted in Nb



Average composition	Atom percent (%)
Nb	70.49
W	3.2
Mo	0.97
Zr	0.98
N	24.34

Fig. 4. Cross-sectional SEM image (a), EPMA elemental maps (b-d) and major elemental concentration profiles (e) of the nitride layer and the substrate of the nitrided Nb521 alloy.

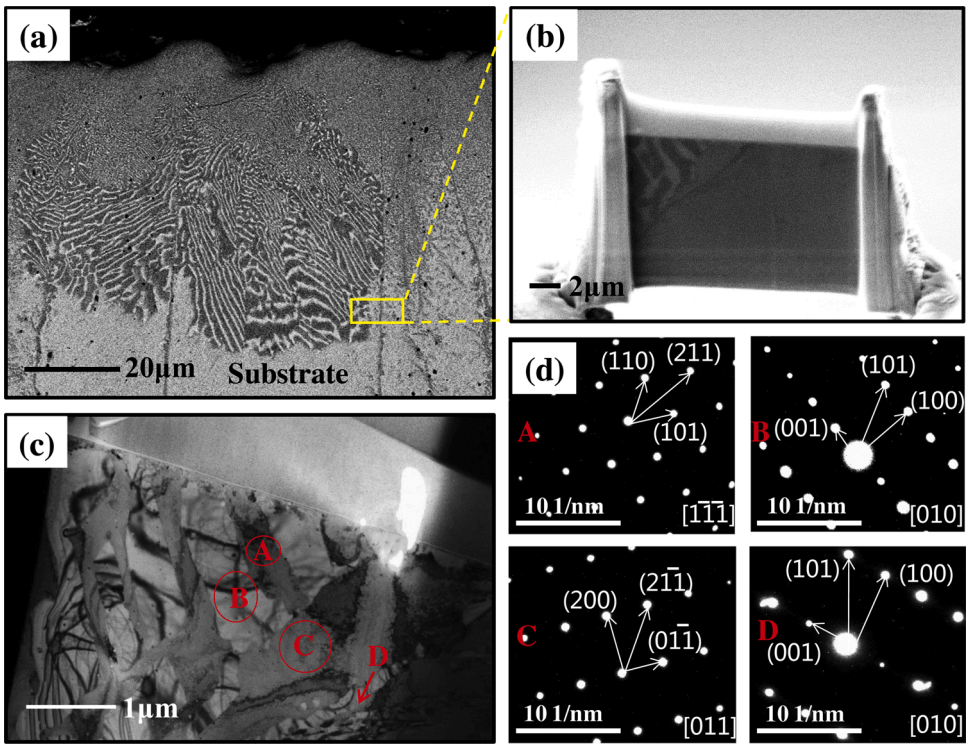


Fig. 5. Backscattered electron image around a LEC zone (a), FIB sample (b) taken from the yellow box in (a), TEM image (c) of the FIB sample and selected area electron diffraction patterns (d) of (c) in the nitrided Nb521 alloy.

Table 1
The composition (at%) at different positions (marked by points) in Fig. 3c.

Positions	Elements				
	Nb	W	Mo	Zr	N
1	67.8	2.4	0.8	1.1	27.9
2	93.8	3.9	1.4	0.9	0
3	66.9	2.2	0.8	1	29.1
4	93.4	4	1.3	1.3	0

but enriched in N (arrows C in Fig. 4d) in comparison to the Nb521 composition, determining the nitrides existing not only in the nitride layer, but also in the grain boundaries. Elemental profiles (Fig. 4e) from surface to substrate (along the red line A in Fig. 4a) suggest that the nitride layer mainly contained elements Nb, W and N, and the average composition of the whole nitride layer was Nb-3.20W-0.97Mo-0.98Zr-24.34 N; thus, Nb_4N_3 and Nb_2N, the constitution phases of the nitride layer (Figs. 1 and 2), are respectively denoted by (Nb, Me)$_2$N and (Nb, Me)$_4$N$_3$ hereafter, where Me represents W, Mo and Zr that dissolved in the nitrides.

Table1 lists the atomic fractions of four special positions, a grain boundary (point 1), a white (point 2) and a black (point 3) phases within the LEC and the substrate (point 4), which are marked in Fig. 3c. The black phase of the LEC zone and the grain boundary contained the N concentration of as high as 27.9~29.1 at.%, preliminarily affirming that they were the (Nb, Me)$_2$N nitride, according to the composition in Table 1. Interestingly, the white phase and the substrate almost showed the same composition and no N was detected within them; therefore, they were the Nb_{SS} phase. To further understand the phase constitution of the grain boundaries and the LEC zones, the region of interest (marked by yellow box in Fig. 5a) was prepared into TEM sample (Fig. 5b) by FIB. In the TEM image of Fig. 5c, the circles A and B respectively represented the black and white phases of the LEC, while the substrate and the grain boundary were denoted by circles C and D, respectively. Selected area electron diffraction (SAED) patterns of the four circles, which are displayed in Fig. 5d, strongly confirmed that both the black phase in the LEC (A in Fig. 5d) and the grain boundary (C in Fig. 5d) were the (Nb, Me)$_2$N nitride with an h.c.p structure, while the white phase (B in Fig. 5d) and the substrate (D in Fig. 5d) were the Nb_{SS} phase. These results derived from the SAEDs were basically in

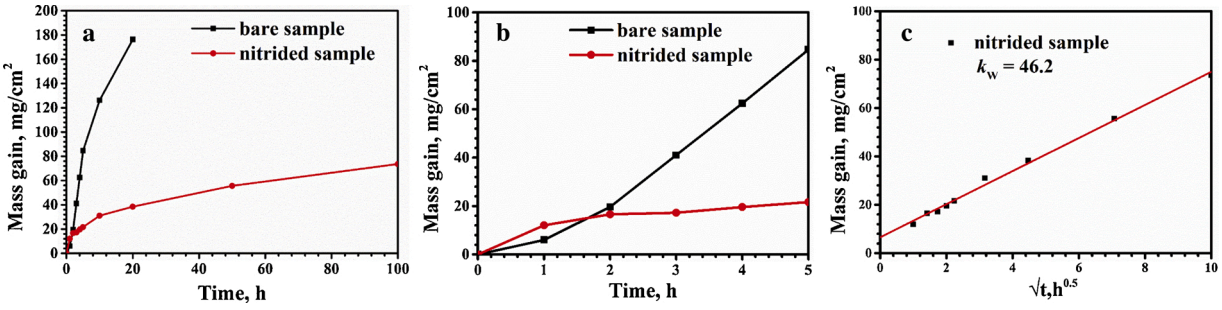


Fig. 6. Oxidation kinetic curves of the bare and nitrided Nb521 samples at 800 °C for 100 h (a) and at beginning of 5 h oxidation (b), and plot of the mass gain (y) vs. the square root of the oxidation time ($t^{1/2}$) of the nitrided Nb521 sample suffered isothermal oxidation (c), where k_W ($mg^2\ cm^{-4}\ h^{-1}$) in (c) is the parabolic rate constant.

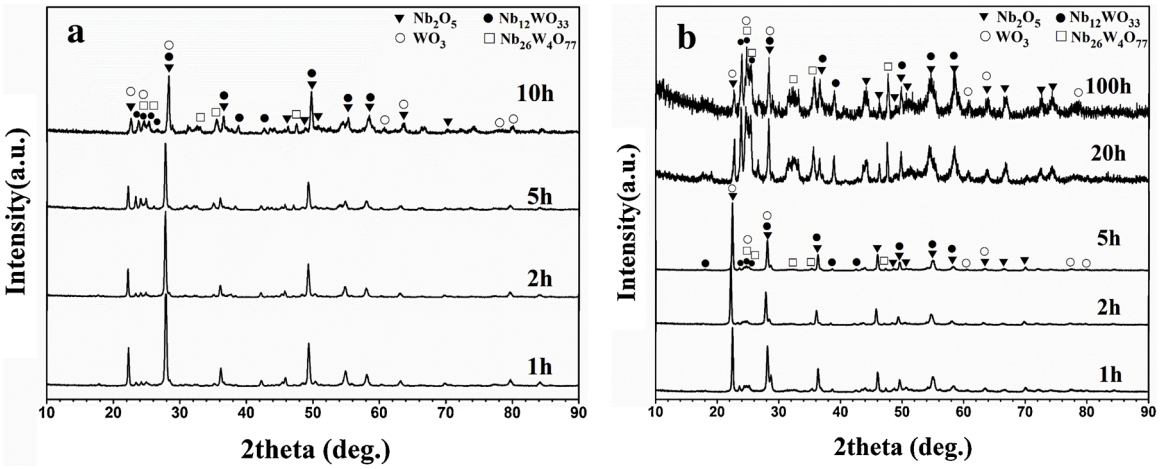


Fig. 7. X-ray diffraction patterns conducted on the Nb521 sample (a) and the nitrided Nb521 sample (b) oxidized at 800 °C for different times.

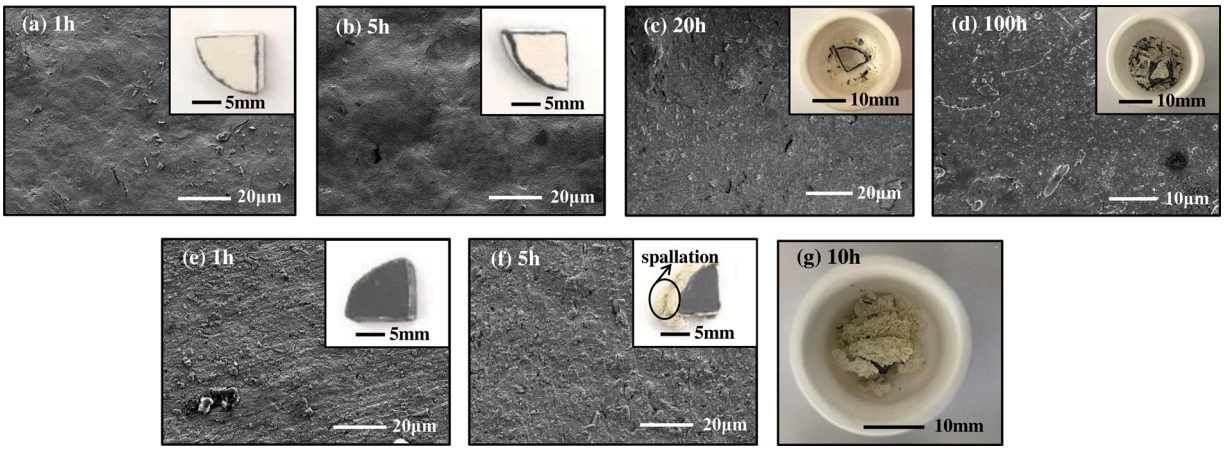


Fig. 8. The surface morphologies of the nitrided samples (a-d) and the bare samples (e-g) oxidized at 800 °C for different times.

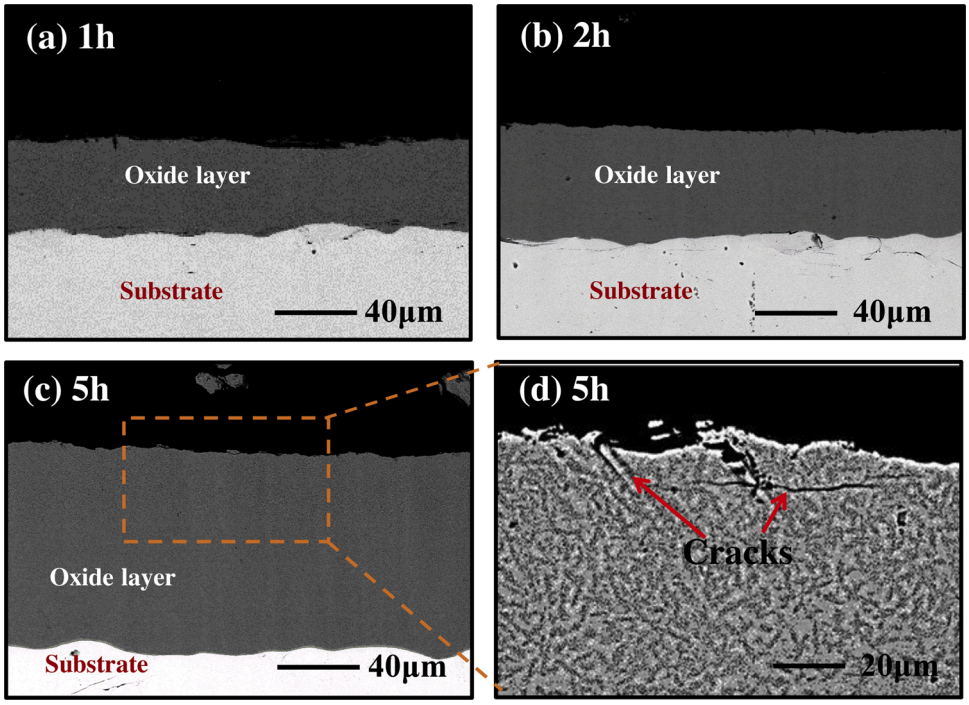


Fig. 9. The cross-sectional BSE images of the bare Nb521 samples oxidized at 800 °C for different times (a-c), and an enlarged image (d) of image (c).

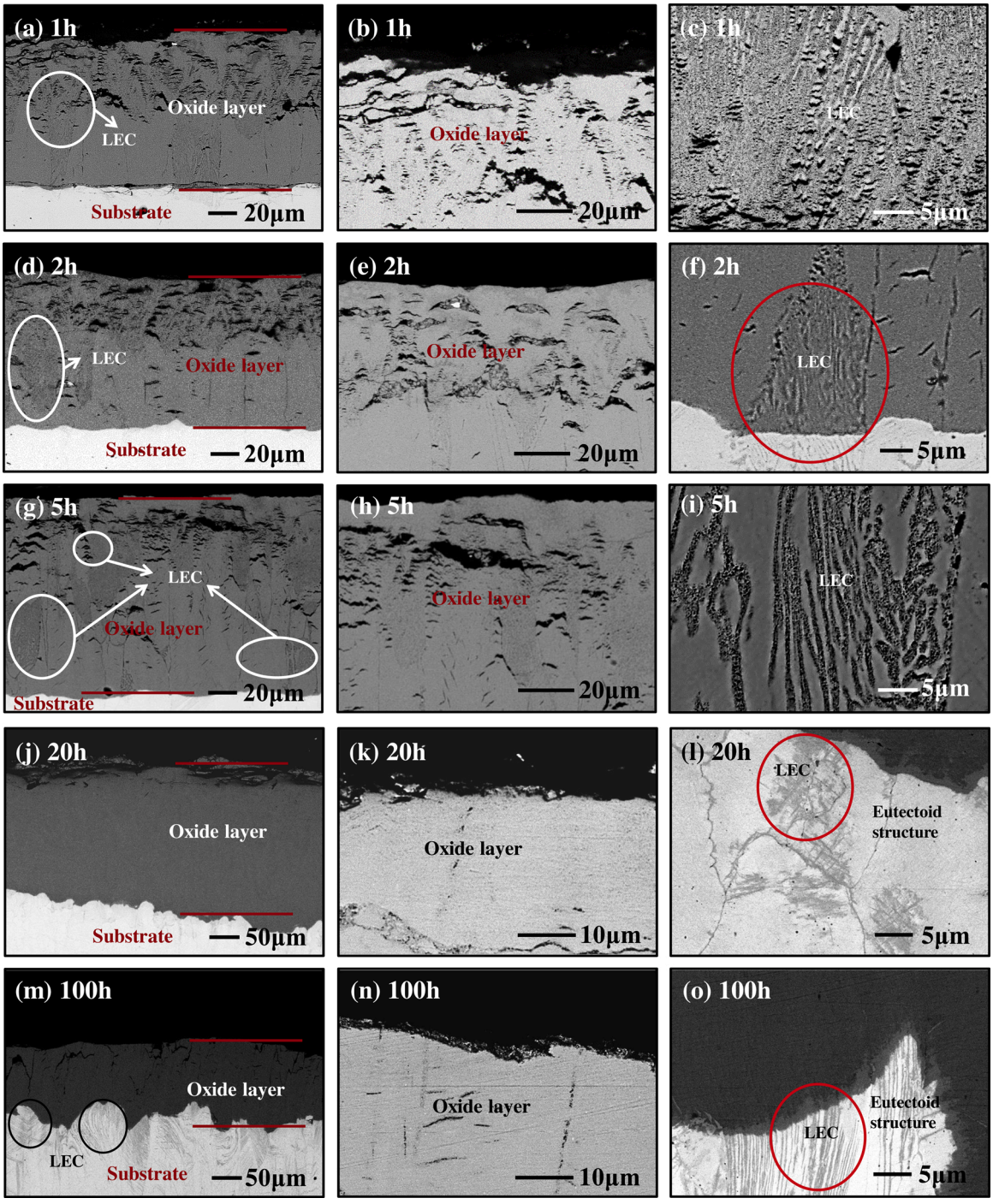


Fig. 10. The Cross-sectional BSE images of the nitrided Nb521 samples oxidized at 800 °C for different times: the oxide layer (a, d, g, j, m), the enlarged oxide layer of upper section (b, e, h, k, n) and the enlarged LEC zones (c, f, i, l, o).

accordance with the XRD patterns (Fig. 1) obtained from the surface of the nitrided samples, except the (Nb, Me)$_4$N$_3$ nitride. The (Nb, Me)$_4$N$_3$ nitride was only detected in the outermost layer of the nitride layer, whereas the interface (containing the LEC zones) of which was composed of the (Nb, Me)$_2$N nitride and the Nb$_{SS}$ phase, but not the (Nb, Me)$_4$N$_3$ nitride. A research [30] has reported that after the nitridation at 1300 °C, the outermost layer of an Nb-10Ti alloy consisted of δ-(Nb,Ti)N$_{1-x}$ nitride, quite different with the (Nb, Me)$_2$N and (Nb, Me)$_4$N$_3$ nitrides formed on the Nb521 alloy. This suggests that the structure of the nitride layer formed on the Nb base alloys depends on the substrate composition.

3.2. Oxidation behaviours of the bare and nitrided Nb521 alloy at 800°C

3.2.1. Oxidation kinetic curves

Fig. 6 shows the oxidation kinetic curves of the bare and nitrided Nb521 samples oxidized in air at 800 °C for 10 h and 100 h, respectively. As can be seen that the nitrided sample had a faster increase in the initial stage of 1 h oxidation (see Fig. 6b) and then the oxidation kinetic curve exhibited a parabolic behavior until oxidation for 100 h (Fig. 6a), whereas the curve for the bare sample almost obeyed a linear law, forecasting a poor oxidation resistance or catastrophic oxidation failure. Note that the test of the bare alloy was stopped when oxidized for 20 h due to almost full "pest" took place at this time (see Fig. 8g). After

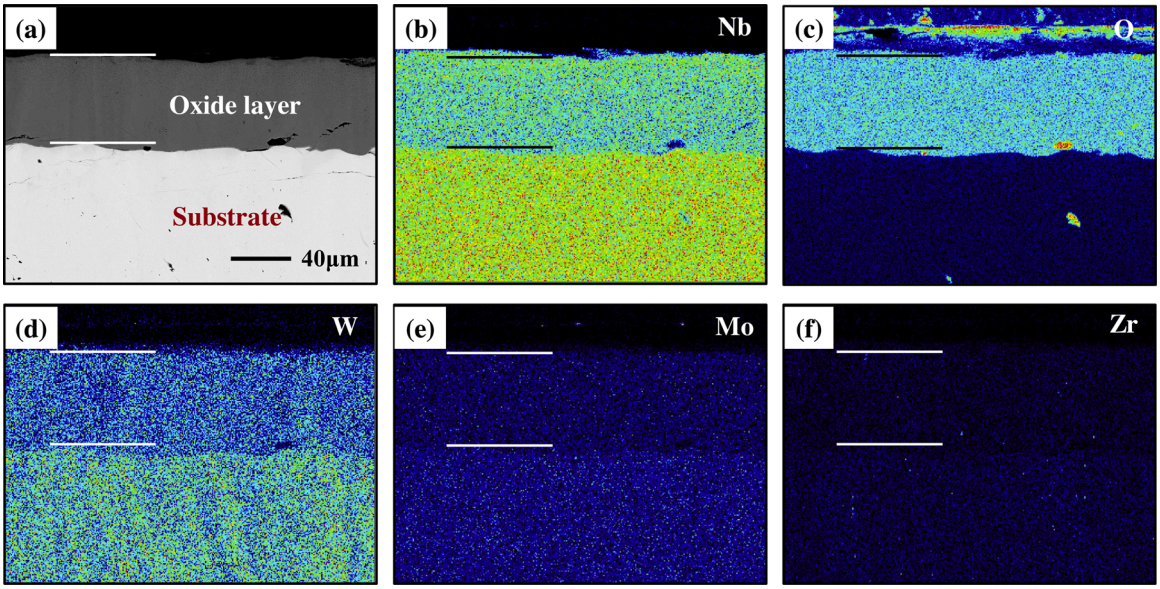


Fig. 11. EPMA elemental maps (b-f) of the oxide layer (a) of the bare Nb521 sample oxidized at 800 °C for 2 h.

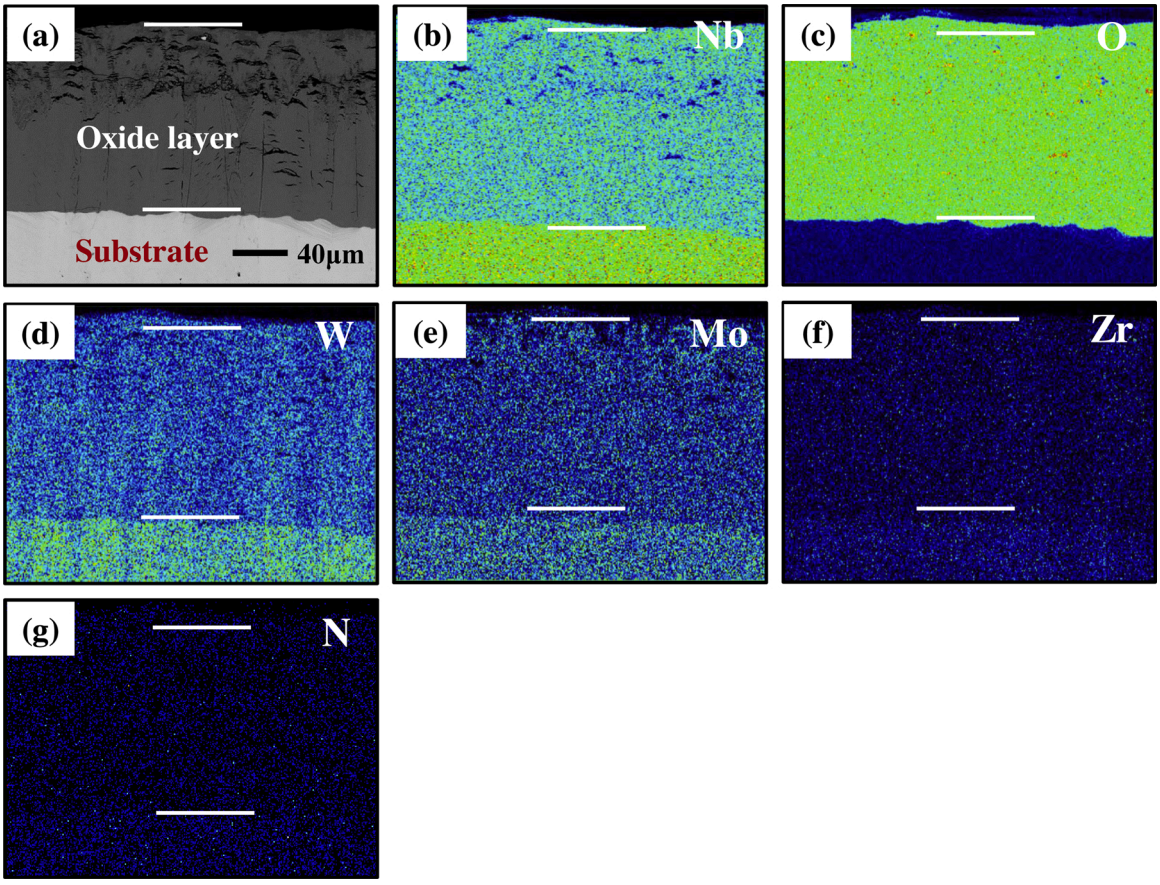


Fig. 12. EPMA elemental maps (b-g) of the oxide layer (a) of the nitrided Nb521 sample oxidized at 800 °C for 2 h.

oxidation at 800 °C for 100 h, the weight gain of the nitrided alloy was 72. 3 mg/cm^2, while that reached as high as 178.5 mg/cm^{-2} for the bare alloy oxidized only for 20 h. The fitted linear relationship between the weight gain (y) and the square root of oxidation time ($\sqrt{t}$) of the nitrided sample is shown in Fig. 6c, from which the parabolic rate constant k_W (46.2 mg^2 cm^{-4} h^{-1}) was calculated according to $y^2 = k_W{\cdot}t$. It is concluded that the nitridation treatment significantly improved the oxidation resistance of the Nb521 alloy.

3.2.2. *Phase constitution and macro-morphologies of the oxide layer*

The XRD patterns detected on the bare and nitrided samples' surfaces oxidized at 800 °C for various times are shown in Fig. 7, respectively. With the similar X-ray profiles of all samples, the oxide products were believed to be independent of the oxide time and the nitride layer. It was

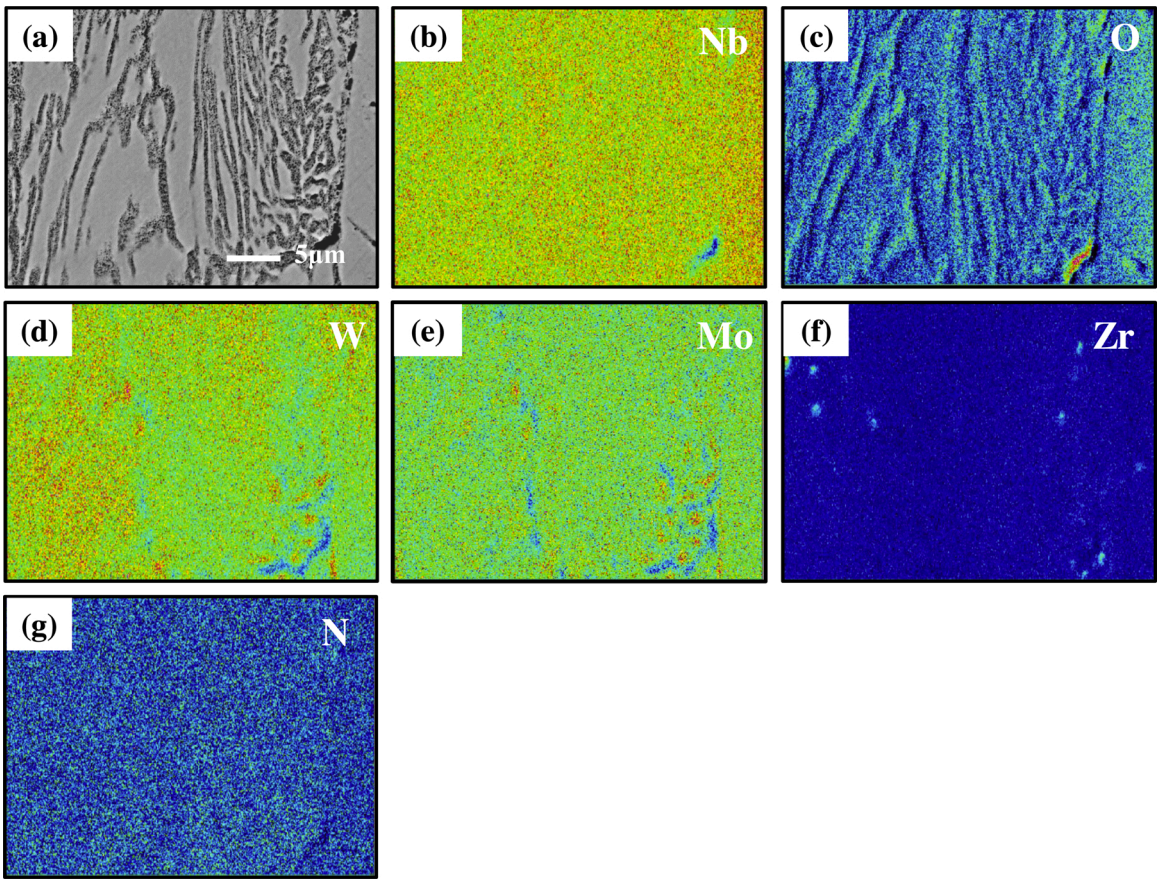


Fig. 13. EPMA elemental maps (b-g) of the LEC zone (a) of the nitrided Nb521 sample oxidized at 800 °C for 5h.

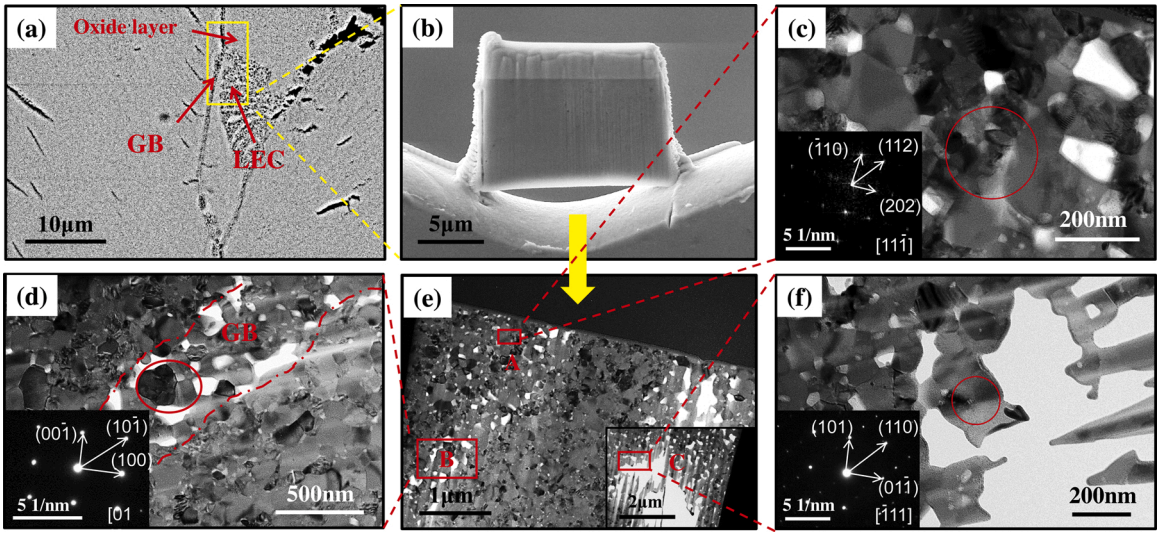


Fig. 14. Backscattered electron image around a LEC zone (a), FIB sample (b) taken from the yellow box in (a), TEM image (e) of the FIB sample and selected area electron diffraction patterns (c, d, f) of (e) in the nitrided Nb521 alloy oxidized at 800 °C for 5 h.

confirmed that the oxide products were most likely composed of Nb_2O_5, WO_3, $Nb_{12}WO_{33}$ and $Nb_{26}W_4O_{77}$ oxides, and the last two complex oxides were the Nb_2O_5 solid solutions, in which the WO_3 oxide was the solute. For convenience, the complex oxides will be referred as to $(Nb, W)_xO_y$ [32], hereafter. A comparison of the macro-morphologies of the bare and nitrided samples after oxidation in air at 800 °C for different times is shown in Fig. 8. The oxidized surface of the nitrided samples was intact and smooth with few defects (Fig. 8a to d) and this oxidized sample almost remained shape resembling the un-oxidized sample before 20 h oxidation, as seen inserts in Fig. 8a to c. When oxidized for 20 h (Fig.8c), slight spallation of the oxide products occurred in the nitrided sample and this phenomenon went on until further oxidation for 100 h without obvious "pest" (Fig. 8d). However, spallation of the oxide products had taken place in the bare sample when oxidized for 5 h (insert in Fig. 8f) and the sample had almost been fully "pest" at 10 h oxidation (Fig. 8g); thus, the fast "pest" failure, corresponding to poor

Table 2
The composition (wt. %) at different positions (marked by boxs) in Fig. 14e.

Position	Elements				
	O	Nb	W	Mo	Zr
A	33.17	54	12.5	0	0.33
B	26.6	73.4	0	0	0
C	25.71	69.99	3.39	0.91	0

oxidation resistance at 800 °C, is the intrinsic behavior of the Nb521 alloy.

3.2.3. *Microstructural evolution of the oxide layer*

Figs. 9 and 10 show the cross-sectional morphologies of the bare and nitrided samples oxidized at 800 °C for different times, respectively, in which distinct microstructures of the oxide layer for the two kinds of samples appeared. For the bare samples, the flat oxide layers grew steadily as the exposure time increased, and the thickness of the remained oxide layers varied from about 35 μm in 1 h (Fig. 9a) to about 95 μm in 5 h (Fig. 9c), note that the spallation of the oxide products (see Fig. 8f) wasn't counted. Cracks that were clear seen in an enlarged image of the box in Fig. 9c, which is shown in Fig. 9d, indicated a further spallation of the oxide layer possibly along the crack direction. As for the nitrided samples shown in Fig. 10, the oxide layers grew rapidly at the initial stage of oxidation and reached 105 μm in thickness after air exposure for 1 h (Fig. 10a) to about 135 μm for 5 h (Fig. 10g). It was worth noting that horizontal but discontinue cracks were frequently observed in the top half of the oxide layer from beginning of the oxidation (see Fig. 10b, e and h), but spallation didn't happen. In the lower part of the oxide layer, the LEC zones still kept the integrality, without notable change in the morphologies, as compared Fig. 10c, f and i to Fig. 3c. When the oxidation time increased to 20 h–100 h, the oxide layer was approximately 150~250 μm in thickness (Fig. 10j and m, spallation of the oxide products in Fig. 8c and d was hard to be counted) and the horizontal cracks could also be seen (Fig. 10k and n). In addition, the LEC zones obviously advanced toward the substrate, as seen circles in Fig. 10l, m and o, suggesting inward diffusion of N during oxidation.

In order to further determine the oxide products formed on the samples, elemental maps of the oxide layer of the bare and nitided samples oxidized at 800 °C for 2 h (stable oxidation stage) were detected and are shown in Figs. 11 and 12, respectively. The EPMA elemental maps revealed that the oxide layer of two samples mainly contained elements Nb, W and O, and its composition was approximately Nb-(0.99~1.05)W-(0.35~0.46)Mo-(0.47~0.49)Zr-(70.10~70.66)O. By combining this composition with the XRD patterns shown in Fig. 7, it is suggested that the oxide product was the (Nb, W)$_x$O$_y$ oxide. The elemental maps of an enlarged image of the LEC oxidized at 800 °C for 5 h shown in Fig. 13c suggested that O had diffused into the LECs.

The phase constitution of the oxide layer of the nitrided sample oxidized for 5 h was further identified through FIB analysis because this oxidized time was in the stable oxidation stage. A TEM sample shown in Fig. 14b was taken from a box in Fig. 14a, in which the oxide layer, grain boundary and part of the LEC zone of the nitrided sample oxidized at 800 °C for 5 h were enclosed. The positions for the TEM observation and the selected area electron diffraction (SAED) are marked in Fig. 14e and the compositions (wt. %) of these positions obtained by EDS are list in Table 2. The TEM image of the grain boundary which is the position B in Fig. 14e is shown in Fig. 14d. Selected area electron diffraction (SAED) pattern (with the beam [0 1 0] of zone axis) inserted in Fig. 14d and EDS analysis (Table 2) confirmed that the oxide product in the grain boundary was mainly the Nb_2O_5 oxide. For the position A within the oxide layer in Fig. 14e, the (Nb,W)$_x$O$_y$ oxide was determined, as analyzed the SAED pattern (with the beam [1 1 -1] of zone axis) inserted in the TEM image of the oxide layer in Fig. 14c) and EDS results. The TEM image of the black region within the LEC zone (position C in Fig. 14e) is shown in Fig. 14f. The insert SAED pattern in Fig. 14f revealed that the oxide product of the black region (the black phase in Fig. 3c) was the (Nb, W)$_2$O$_5$ oxide. For the Nb_{SS} phase in the LEC (the white phase in Fig. 3c), the oxide product should be (Nb, W)$_x$O$_y$, the same as that of the oxidized bare sample with the single Nb_{SS} microstructure.

4. Discussions

4.1. *Formation of nitride layer on the Nb521 alloy*

In transition metal nitides, nitrogen diffuses much faster than metal [28]; therefore, the generation of the nitrides is controlled by the inward

Table 3
The standard Gibbs free energies (ΔG°, kJ/mol) and the minimum values of the oxygen partial pressures for the reactions (3)-(7) at 800 °C.

Reactions	Temperature	Standard Gibbs free energy (ΔG°)	Minimum oxygen partial pressure (Pa)
3		−572.782 KJ/mol	1.30×10^{-28}
4		−378.604 KJ/mol	3.70×10^{-19}
5	800 °C	−317.202 KJ/mol	3.61×10^{-16}
6		−896.104 KJ/mol	2.37×10^{-44}
7		−336.508 KJ/mol	4.18×10^{-17}

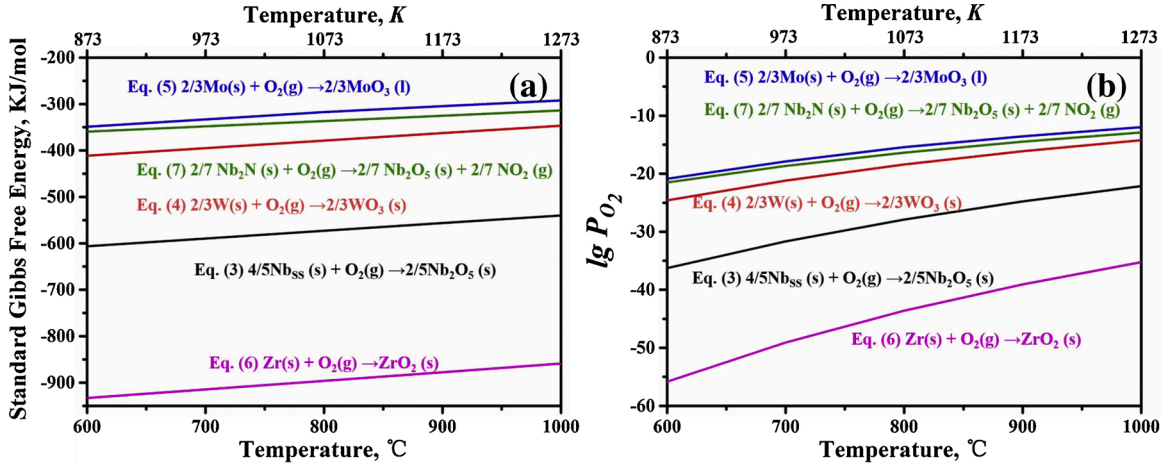


Fig. 15. Variation of the standard Gibbs free energies (ΔG°, kJ/mol) (a) and the minimum values of the oxygen partial pressures (P_{O2}) (b) for the Eqs. (3)–(7) with temperature.

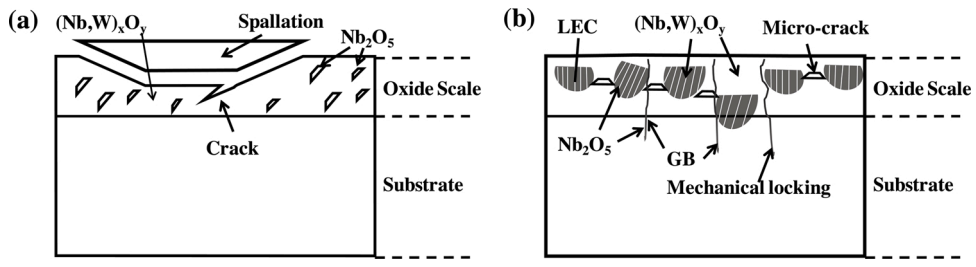


Fig. 16. The oxidation mechanisms of the bare (a) and nitrided (b) Nb521 alloy.

diffusion of nitrogen from air. In general, the nitridation processing of the Nb521 alloy can be divided into three steps. Firstly, nitrogen in the atmosphere reacts with the Nb521 alloy and forms nitrides on the sample surface. The outermost nitride layer composes of (Nb, Me)$_4$N$_3$ and (Nb, Me)$_2$N formed via Eqs. (1) and (2), where the two nitrides have been detected by XRD (Fig. 1).

$$N_2(g) + 8/3Nb_{SS}(s) \rightarrow 2/3(Nb, Me)_4N_3\ (s) \quad (1)$$

$$N_2(g) + 4Nb_{SS}(s) \rightarrow 2(Nb, Me)_2N(s) \quad (2)$$

Secondly, nitrogen continues to inward diffuse form the nitride layer, subsequently, the nitride layer becomes thicker; at this moment nitrogen diffuses in the grain boundaries faster than in grain interiors, leading to the preferential formation of the nitride in the grain boundaries (Fig. 4a and d). Finally, nitrogen diffuses from the interface between the nitride layer and the substrate and the grain boundaries to produce the needle-like nitrides (Fig. 3b) or the LECs composed of (Nb, Me)$_2$N and Nb$_{SS}$ phases (Fig. 3a and c). A similar phenomenon has been observed in the nitrided Nb-10 at.% Ti alloy [19]. According to Nb-N binary phase diagram [33], the critical amount for the formation of (Nb, Me)$_4$N$_3$ is 40 at %, however, the N content in the LEC zone is just 29.1 at.% (Table 1), which is not enough to support reaction of Eq. (1) and formation of the (Nb, Me)$_4$N$_3$ phase in the LEC zone. For this reason, the reaction Eq. (2) takes place and the (Nb, Me)$_2$N and Nb$_{SS}$ phases exists in the LEC zones (Fig. 5).

4.2. Oxidation mechanisms

For the Nb521 alloy with single Nb$_{SS}$ phase, each element reacts separately with O. At 800 °C of the onset oxidation, Nb reacts with O to form Nb$_2$O$_5$ as following Eq. (3) as the oxygen diffuses inward,

$$4/5Nb_{SS}\ (s) + O_2(g) \rightarrow 2/5Nb_2O_5\ (s) \quad (3)$$

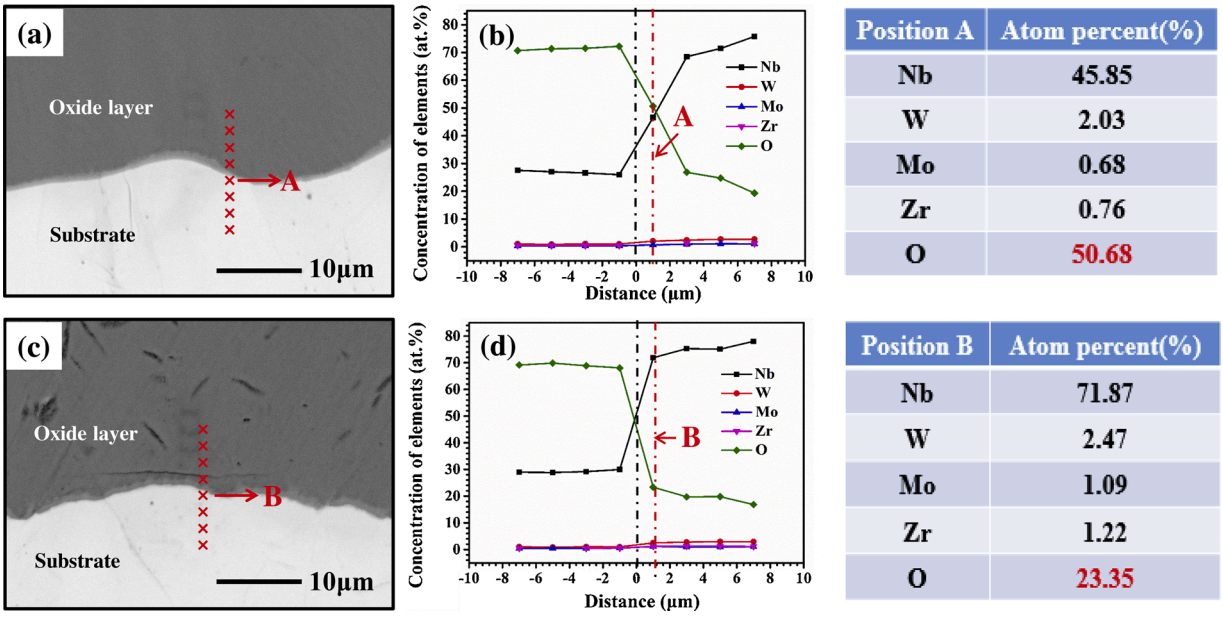
Meanwhile, the atomic W, Mo and Zr solidly dissolved in the Nb$_{SS}$ phase are oxidized through the Eqs. (4), (5) and (6):

$$2/3W(s) + O_2(g) \rightarrow 2/3WO_3\ (s) \quad (4)$$

$$2/3Mo(s) + O_2(g) \rightarrow 2/3MoO_3\ (l) \quad (5)$$

$$Zr(s) + O_2(g) \rightarrow ZrO_2\ (s) \quad (6)$$

Fig. 15 shows the standard Gibbs free energies (ΔG°) and the minimum values of the oxygen partial pressure (P_{O2}) of the elements in the bare Nb521 alloy dependent on temperatures, while Table 3 lists the ΔG° and the P_{O2} values to enable occurring of the Eqs. (3)~(7) reactions at 800 °C. Because the MoO$_3$ is liquid at 800 °C and the Zr content is as low as 1 at.%, the oxide products we detected are mainly the Nb$_2$O$_5$, WO$_3$ and (Nb$_2$O$_5$+WO$_3$) solid solution (Nb,W)$_x$O$_y$ via XRD (Fig. 7). From the ΔG° and the P_{O2} values shown in Table 3, Nb is deducted to be oxidized preferentially to form the Nb$_2$O$_5$ oxide in the initial stage of oxidation; then the WO$_3$ oxide forms resulted from the reaction of W to O, and finally, the Nb$_2$O$_5$ and WO$_3$ oxides dissolve with each other to



Position A	Atom percent(%)
Nb	45.85
W	2.03
Mo	0.68
Zr	0.76
O	50.68

Position B	Atom percent(%)
Nb	71.87
W	2.47
Mo	1.09
Zr	1.22
O	23.35

Fig. 17. Cross-sectional SEM images (a, c) and major elemental concentration profiles (b, d) from the oxide layer to the substrate of the bare (a, b) and nitrided Nb521 (c, d) samples after oxidized at 800 °C for 2 h.

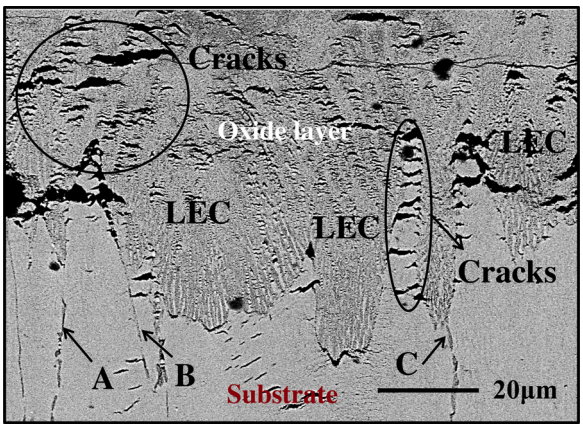


Fig. 18. The Cross-sectional BSE image of the oxide layer on the nitrided sample oxidized at 800 °C for 5h.

generate the $(Nb,W)_xO_y$ oxide. The oxidation mechanism of the bare Nb521 alloy is shown schematically in Fig. 16a.

When the Nb521 alloy has a nitride layer, the nitrides may react with O following the Eqs. (7) and (8)

$$2/7\ Nb_2N\ (s) + O_2(g) \rightarrow 2/7\ Nb_2O_5\ (s) + 2/7\ NO_2\ (g) \quad (7)$$

$$1/8\ Nb_4N_3\ (s) + O_2(g) \rightarrow 1/4\ Nb_2O_5\ (s) + 3/8\ NO_2\ (g) \quad (8)$$

Unfortunately, the standard Gibbs free energy of Eq. (8) is unavailable; thus, we cannot determine which one of the reactions of Nb_4N_3 or Nb_2N to O is dominant right now. Anyway, the whole $(Nb,W)_xO_y$ products detected in the original nitride layer without any Nb_4N_3 or Nb_2N remaining strongly emphasize that all the nitrides have participated in the oxidation. A schematic diagram with regard to oxidation of the nitride layer on the Nb521 alloy is shown in Fig. 16b.

4.3. Mechanism of the nitride layer preventing oxidation

It can be seen from Figs. 6 and 8 that the nitrided Nb521 sample shows better oxidation resistance than the bare sample. Table 3 still summarizes the standard Gibbs free energy of the Nb_2N reacting with O for comparison. More energy and greater O content (corresponding to the higher O partial pressure, see Table 3) are needed to meet the requirement of the reaction of the Nb_2N with O (Eq. (7)) rather than those of the dominant elements Nb and W with O (Eqs. (3) and (4)). Therefore, the nitride layer cannot be oxidized easily, which indicates that the nitrides are more stable when at the atmosphere and high temperatures, as compared with the Nb_{SS} alloy. This is the first factor why the nitridation treatment is an effective technique to prevent the Nb521 substrate from attack of O in the atmosphere.

Secondly, the nitride layer may further impede O diffusing toward the substrate to suppress the oxidation. Fig. 17b and d show the elemental concentration profiles at various depths in the cross-sectional images of Fig.17a and c of the bare and nitrided Nb521 samples after oxidation for 2 h, respectively. In order to obtain a more accurate composition, five positions at the same depth were selected and detected by EPMA, and naturally, the O content decreases with an increase of the depth. Comparison reveals that for any position with the same distance from the front of the oxide layer, the O content in the nitrided sample is always less than that in the bare sample. For instance, at the point just beneath the oxide scale (position A in Fig. 17a) the O content is approximately 50.68 at.% for the bare sample and that is 23.35 at.% (position B in Fig. 17c) for the nitrided sample. This suggests that the formation of the nitride layer can reduce the rate of O diffusion into the substrate, thereby improving the oxidation resistance of the Nb521 alloy. It should be emphasized that this mechanism is also available in the grain boundaries (GBs) because of existence of the nitrides in the grain boundaries. The nitrides in the grain boundaries may also play a significantly role in impeding the diffusion of O or the oxidation, because the diffusion of O along the grain boundary is quite faster than that along the grain interior.

Finally, the nitrides in the grain boundaries and the LEC zones may act as "mechanical locking" to avoid spallation of the oxide products at early stage of oxidation and "pest" at long time oxidation (Figs. 8 and 10). As shown in Fig. 18, the nitrides in the grain boundaries (A, B, C in Fig. 18) and the LEC zones obviously connect with the oxide layer, without "pest" and spallation of the oxide layer in the nitrided samples (Fig. 8b) when oxidized for 5 h even though short cracks parallel to the sample surface frequently arising within the oxide layer (Figs. 10b, e, h and 18). This phenomenon reveals that the nitrides hinder diffusion of O toward the front of the horizontal crack tip (proved by lower O content in the nitride layer, see Tables in Fig. 17) and postpones further propagation of the crack and spallation of the oxide products enforced by oxidation until oxidation for 20 h (Fig. 8c). Combining the aforementioned mechanisms of the nitride layer and the role of "mechanical locking" of the nitrides in the grain boundaries (Fig. 18) and the LECs (Fig. 10l, m and o), we can conclude that the "pest" of the oxide products is significantly suppressed by the nitrides (Fig. 8d), and the nitridation is beneficial to improve the oxidation resistance of the Nb521 alloy. For the bare sample, the phenomena that heavy spallation of the oxide products (Fig. 8f), crack continuous propagation toward the substrate (Fig. 9d), as well as full "pest" at oxidation for 10 h (Fig. 8g), strongly confirm much poor oxidation resistance of the bare sample.

5. Conclusions

A comparative study on the microstructures and the oxidation behaviours of a bare and nitrided Nb_{SS} alloy (Nb521) at moderate temperatures of 800 °C was conducted. The major conclusions are summarized as follows.

1) A compact nitride layer with a thickness of about 10 μm was prepared on the Nb_{SS} alloy by gas nitridation treatment at 1200 °C for 12 h. The outermost nitride layer was composed of the $(Nb, Me)_4N_3$ and $(Nb, Me)_2N$ nitrides, while the lamellar eutectoid-like colonies (LECs) containing the $(Nb, Me)_2N$ and Nb_{SS} phases were formed just beneath the outermost nitride layer.
2) After oxidation at 800 °C for 100 h, the kinetic curve of the nitrided alloy followed a parabolic law, while that of the bare sample showed a linear law. The mass gain of the nitrided sample was 72.3 mg/cm^2 when oxidized for 100 h, while that of the bare sample oxidized for 20 h was 178.5 mg/cm^2. The oxide layer of the nitrided sample always remained intact without obvious "pest" in whole oxidation period, whereas for the bare sample full "pest" of the oxide layer took place when oxidized for 10 h.
3) The oxide layer of the nitride sample consisted of the outermost $(Nb, W)_xO_y$ layer and the inner $(Nb_2O_5 + (Nb,W)_xO_y)$ zones, where the later formed on the original LEC zones. Impeding internal diffusion of O by the nitrided layer and LECs and "mechanical locking" of the nitrides formed in the grain boundaries and the LECs were the mechanisms for the nitridation processing to improve the oxidation resistance of the Nb521 alloy at the moderate temperatures.

Declaration of Competing Interest

The authors declared that there is no conflict of interest to this work.

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